

How hard can a black hole **spin**?

Extremality shift of rotating black holes
at finite Gauss–Bonnet coupling

C. Bejarano D. Blanco A. Goya G. Pérez Nadal
with **Claude Code** and **Codex**

AI agents course (M. Zaldarriaga & N. Mirón Granese)

FCEN–UBA · 2 October 2026 · github.com/ddblanco/BHs



THE QUESTION

There is a speed limit.

Kerr (4D, GR): $M^2 \geq |J|$

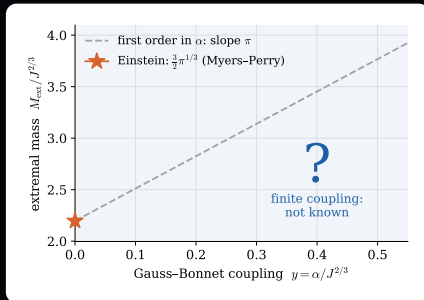
Myers–Perry (5D, GR): $M \geq \frac{3}{2} \pi^{1/3} J^{2/3}$

$$\mathcal{L} = R + 2\Lambda + \alpha (R^2 - 4R_{ab}^2 + R_{abcd}^2)$$

Add Gauss–Bonnet, coupling α : it moves.

Known only for tiny α : slope = π .

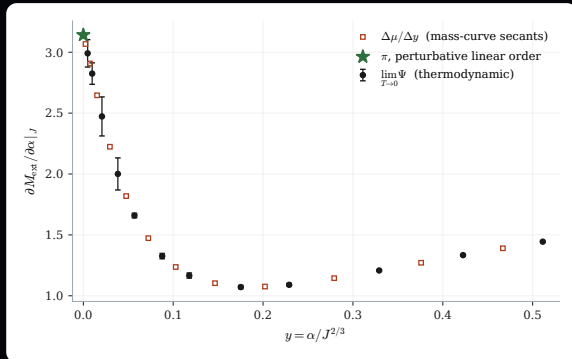
And at **finite** α ?



THE RESULT

The limit gets **tighter**.

But not the way first order said.



extended first law

$$dM = T dS + 2 \Omega_H dJ + \boxed{\Psi d\alpha}$$

at $T = 0$, fixed J :

$$\Psi_{\text{ext}} = \left. \frac{\partial M_{\text{ext}}}{\partial \alpha} \right|_J$$

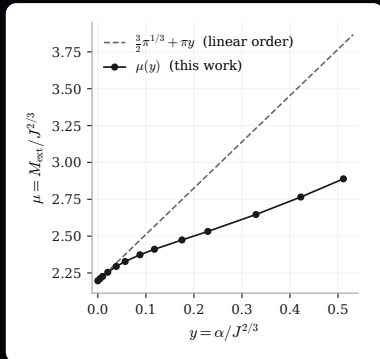
$$\pi \rightarrow 1.07 \rightarrow 1.44$$

slope $\partial M_{\text{ext}}/\partial\alpha|_J$: falls, then rises

THE RESULT

The limit gets **tighter**.

But not the way first order said.



extended first law

$$dM = T dS + 2 \Omega_H dJ + \boxed{\Psi d\alpha}$$

at $T = 0$, fixed J :

$$\Psi_{\text{ext}} = \left. \frac{\partial M_{\text{ext}}}{\partial \alpha} \right|_J$$

+31%

growth of the extremal mass M_{ext}

2.3×

first order overshoots the real shift

WHY BELIEVE IT

10

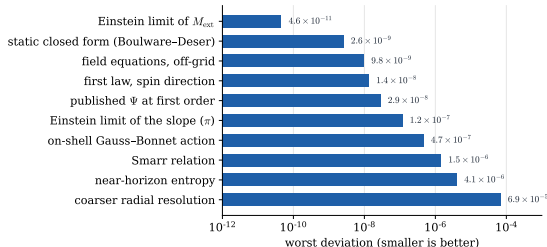
independent checks.

All below 10^{-4} .

Known solutions first: arXiv:1010.0860
to 2×10^{-4} .

Two other formulas for the slope (Smarr relation, Gauss-Bonnet action) agree with our derivative method to $\sim 10^{-6}$.

Largest uncertainty: the $T \rightarrow 0$ extrapolation.



WHY BELIEVE IT

10

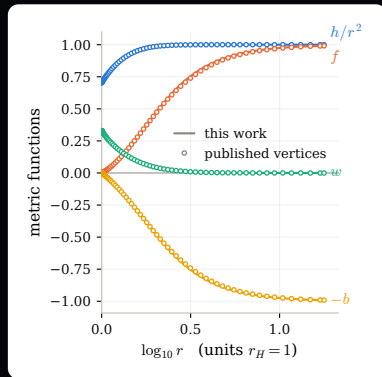
independent checks.

All below 10^{-4} .

Known solutions first: arXiv:1010.0860
to 2×10^{-4} .

Two other formulas for the slope (Smarr
relation, Gauss–Bonnet action) agree
with our derivative method to $\sim 10^{-6}$.

Largest uncertainty: the $T \rightarrow 0$ extrapolation.



lines: our solver circles: arXiv:1010.0860, fig. 1

WHY BELIEVE IT

10

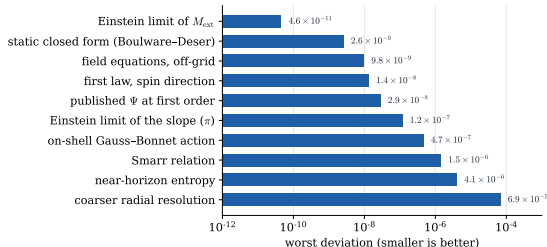
independent checks.

All below 10^{-4} .

Known solutions first: arXiv:1010.0860
to 2×10^{-4} .

Two other formulas for the slope (Smarr
relation, Gauss–Bonnet action) agree
with our derivative method to $\sim 10^{-6}$.

Largest uncertainty: the $T \rightarrow 0$ extrapolation.



HOW IT WAS MADE & WHAT WE LEARNED

Two AI agents. One of them the **referee**.

36.8 h

agent time, 13 days

1.42 B

tokens processed

79

human prompts

(4.4 h reproducing, 11.2 h new solver, 8.7 h $\Psi_{T \rightarrow 0}$)

Writing the code was cheap.

Verifying it was the work.

The background of the slide is dark with a central, glowing orange and yellow pixelated shape. Scattered around this central shape are several faint, light-colored diagrams of neural networks, each consisting of nodes connected by lines. Additionally, there are numerous binary digits (0s and 1s) floating in the background, some appearing to be part of the neural network structures.

Thank you for your attention

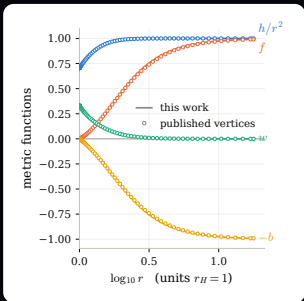
Matías and Nahuel, thanks for the AI course

A1 · Numerical method: solving the field equations

Equal-spin ansatz $\Rightarrow$ ODEs in r for b, f, h, w :

$$ds^2 = -b dt^2 + \frac{dr^2}{f} + r^2 \left[d\theta^2 + \sin^2\theta \cos^2\theta (d\varphi_1 - d\varphi_2)^2 \right] \\ + h \left[\sin^2\theta d\varphi_1 + \cos^2\theta d\varphi_2 - w dt \right]^2$$

horizon: $b = f = 0$, $w = \Omega_H$; infinity: $b, f, h/r^2 \rightarrow 1$, $w \rightarrow 0$



Discretise: $\chi = 1 - r_H/r$, Chebyshev–Lobatto,

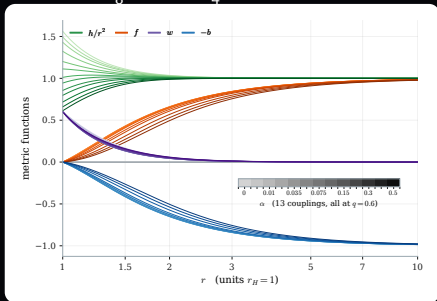
$N = 64$: $R_N(u; \alpha, r_H, \Omega_H) = 0$ ($4N$ eqs.)

Solve: damped Newton; continuation from Myers–Perry

Accept: $G_{\mu\nu} + \alpha H_{\mu\nu} < 10^{-8}$ at 51 off-grid points

Read off: $b \simeq 1 + U/r^2$, $w \simeq W/r^4 \Rightarrow$

$$M = -\frac{3\pi}{8} U, \quad J = \frac{\pi}{4} W$$



A2 · Numerical method: measuring Ψ and reaching $T = 0$

Ψ from derivatives at fixed (r_H, Ω_H)

$$\Psi = \left. \frac{\partial M}{\partial \alpha} \right|_{S,J} = M_\alpha - T S_\alpha - 2\Omega_H J_\alpha$$

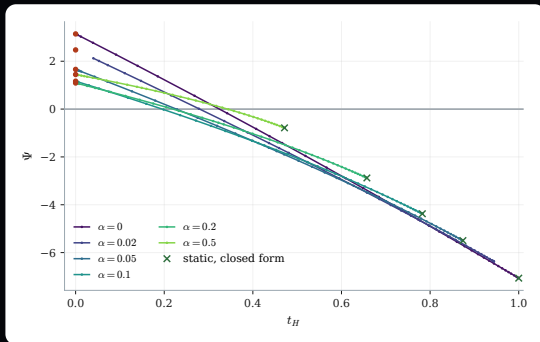
Derivatives without finite differences

$$\underbrace{\frac{\partial R_N}{\partial u}}_{\text{Newton Jacobian}} \frac{du}{d\alpha} = - \frac{\partial R_N}{\partial \alpha}$$

one linear solve, no step size (implicit function theorem)

Get cold, then extrapolate

cubic fit through the 6 coldest states;
coldest $TJ^{1/3} \approx 3 \times 10^{-6}$



Ψ against temperature; red dots: the $T = 0$ values (the slope)

A2b · Two other routes to Ψ

1. Smarr relation (scaling)

weights: $M, \alpha \rightarrow 2$, $S, J \rightarrow 3$; Euler's theorem:

$$2M = 3TS + 6\Omega_H J + 2\alpha\Psi$$
$$\Rightarrow \Psi = \frac{2M - 3TS - 6\Omega_H J}{2\alpha}$$

charges of **one** solution: no Jacobian, no derivative
agrees to 1.45×10^{-6} (313 states); useless as $\alpha \rightarrow 0$

2. On-shell Gauss–Bonnet action

Gibbs potential = on-shell action per unit time; at fixed (T, Ω_H) , α enters explicitly only through $\mathcal{L}_{\text{GB}}$ (field equations remove the rest):

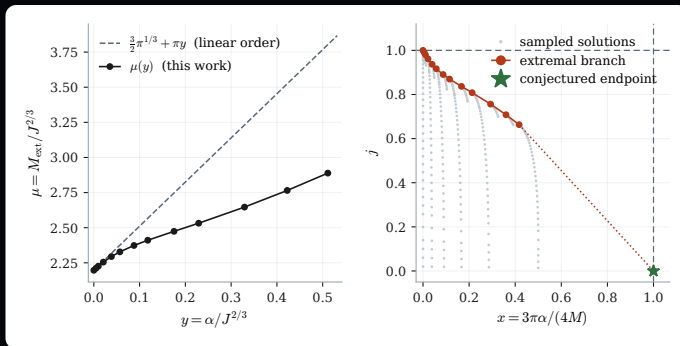
$$\Psi = -\frac{1}{16\pi} \int_{\Sigma} \sqrt{-g} \mathcal{L}_{\text{GB}}$$

curvature of the solution only: no charge extraction
either

agrees to 4.67×10^{-7} (43 states, away from extremality)

Both use the same numerical solutions; neither uses the linear solve of A2.

A3 · Physical meaning of the result



Spin bound

$\Psi_{\text{ext}} > 0 \Leftrightarrow j_{\text{ext}}$ falls:
less spin per unit mass

Slope minimum

$$\begin{aligned} 3\omega &= \mu - y\mu' \\ &= \text{fastest extremal horizon,} \\ \omega &= \Omega_H J^{1/3} \end{aligned}$$

Why it rises again

$$\begin{aligned} M &> \frac{3\pi}{4}\alpha \text{ forces} \\ \mu' &\rightarrow \frac{3\pi}{4} \approx 2.36 \end{aligned}$$

A4 · How it started: the first prompts

“El proyecto consiste en utilizar agentes de IA para implementar códigos numéricos que permitan obtener soluciones de agujeros negros rotantes en dimensión $D > 4$ [...] reproducir soluciones numéricas ya encontradas en la literatura. [...] el paso siguiente sería emprender la búsqueda de nuevas soluciones.”

Prompt 1 — 4 Sept, 19:16, to Codex · then “Fijate que el proyecto realmente siga las indicaciones propuestas por el docente del curso” (3), “ejecutar con subagentes” (5), “continuemos con el hito 3”, “dale, avanza” · 79 human prompts, 4–28 Sept · Claude Code builds ↔ Codex referees

```
ruta-de-trabajo-inicial.txt 5 Sept 2026
52 ## Hito 4 – EGB rotante asintóticamente plano
53
54 *Salida mínima exitosa del proyecto:* familia parcial, pero validada, de
55 soluciones D=5 con momentos angulares iguales que reproduzca resultados de
56 Brihaye, Kleihaus, Kunz y Radu (2010).
57
58 - Continuar primero en velocidad angular y luego en alpha.
59 - Comparar magnitudes adimensionales y curvas publicadas.
60 - Comprobar regularidad, cargas, temperatura, entropía y primera ley.
61 - Documentar zonas donde el método pierde convergencia sin interpretarlas de
62 inmediato como fronteras físicas.
63 ... (Hito 5 – método híbrido asistido por IA)
76 ## Hito 6 – Exploración nueva
77
78 *Salida ambiciosa:* evidencia numérica reproducible sobre una extensión no
79 cubierta por los benchmarks.
80
81 Sólo se seleccionará después del Hito 4. Candidatos: otra densidad de Lovelock,
82 otra condición asintótica o una región no explorada del espacio de parámetros.
83 El spin desigual o un ansatz de cohomogeneidad 2 se consideran proyectos de
84 segunda generación por requerir PDEs.
```

first roadmap, 5 Sept. Hito 6 became Ψ ; Hito 8 ($T \rightarrow 0$, paper, referees) was not in the plan.

4.1.3 Thermodynamical properties

Considering the thermodynamics of these solutions, the EGB black holes should satisfy the first law of thermodynamics

$$dE = T_H dS + 2\Omega_H dJ. \quad (4.3)$$

One may regard the parameters S, J as a complete set of extensive parameters for the mass-energy $E(S, J)$ and define the intensive parameters conjugate to them. These quantities are the temperature and the angular velocities. Also, for $\alpha \neq 0$, these solutions do not appear to satisfy any simple Smarr relation¹⁴.

¹⁴The Smarr relation in Einstein gravity is $E = 3(T_H S + 2\Omega_H J)/2$. The recent work [43] proposes a Smarr relation for static black holes in Lovelock gravity. It would be interesting to extend the formalism in [43] to the case of spinning solutions.

arXiv:1010.0860, §4.1.3 and footnote 14.

Hito 6, α as a thermodynamic variable:

$$2M = 3TS + 6\Omega_H J + 2\alpha\Psi$$

A5 · What it cost

36.8 h

active agent time, 13 days

1.42 billion

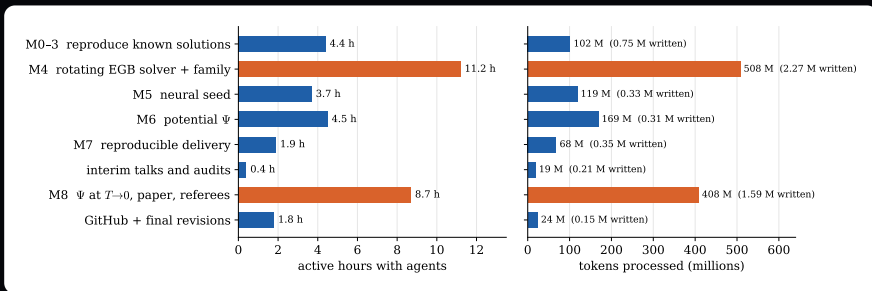
tokens processed
(99% re-read context)

5.9 million

tokens written (≈ 24 MB)

79

human prompts



The first **new** object cost $5\times$ the tokens of reproducing everything known.

Not counted: human time, background numerical runs.